LeccionClusteringJerarquico

November 4, 2026

[5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
```

[18]:
```
data = pd.read_csv('Wholesale customers data.csv')
data.head()
```

[18]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	7844
3	1	3	13265	1196	4221	6404	507	1788
4	2	3	22615	5410	7198	3915	1777	5185

[20]:
```
from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

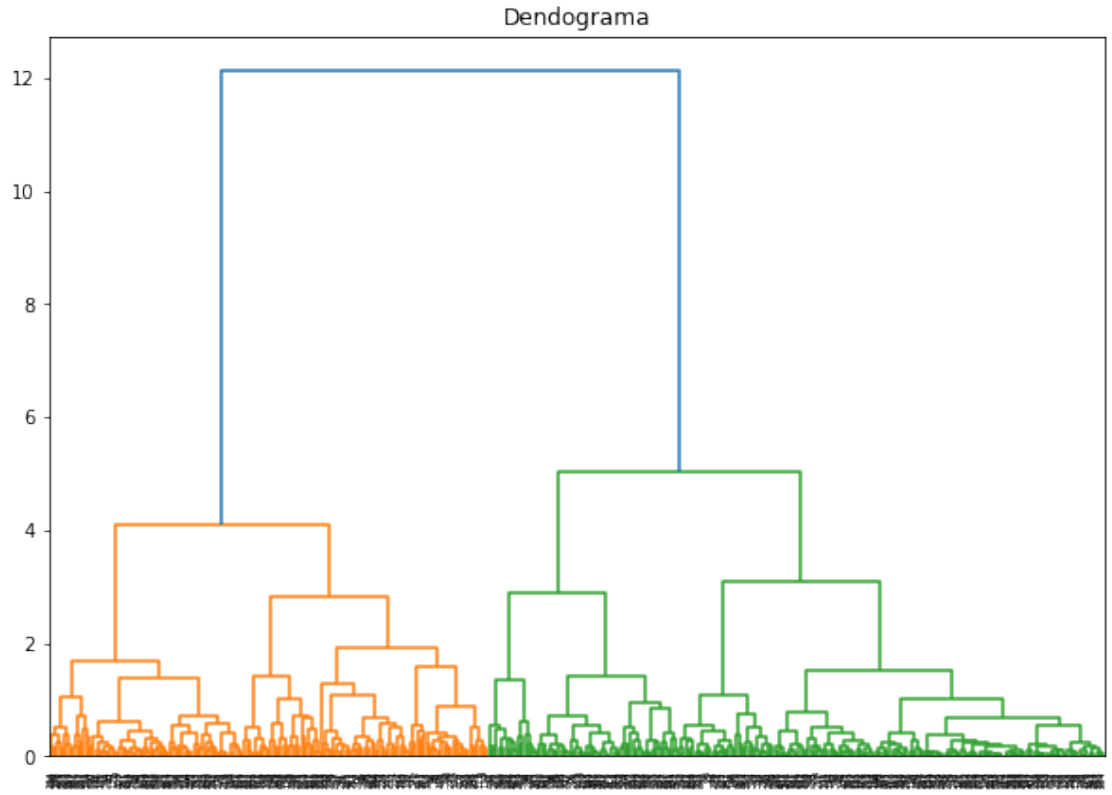
[20]:
```
array([[1.11821406e-04, 1.67732109e-04, 7.08332695e-01, …,
        1.19648904e-02, 1.49505220e-01, 7.48085205e-02],
       [1.25321880e-04, 1.87982820e-04, 4.42198253e-01, …,
        1.10408576e-01, 2.06342475e-01, 1.11285829e-01],
       [1.24839188e-04, 1.87258782e-04, 3.96551681e-01, …,
        1.50119124e-01, 2.19467293e-01, 4.89619296e-01],
       …,
       [5.01633106e-05, 7.52449659e-05, 3.64461533e-01, …,
        1.09606834e-02, 3.72236846e-01, 4.68274505e-02],
       [9.11309417e-05, 2.73392825e-04, 9.37737390e-01, …,
        9.45939175e-02, 1.53099982e-02, 1.93653251e-01],
       [2.41225630e-04, 7.23676891e-04, 6.72295832e-01, …,
        1.56796660e-02, 1.15064626e-01, 1.25437328e-02]])
```

[22]:
```
data_scaled = pd.DataFrame(data_scaled, columns = data.columns)
data_scaled.head()
```

[22]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	0.000112	0.000168	0.708333	0.539874	0.422741	0.011965	0.149505	0.074809
1	0.000125	0.000188	0.442198	0.614704	0.599540	0.110409	0.206342	0.111286
2	0.000125	0.000187	0.396552	0.549792	0.479632	0.150119	0.219467	0.489619
3	0.000065	0.000194	0.856837	0.077254	0.272650	0.413659	0.032749	0.115494
4	0.000079	0.000119	0.895416	0.214203	0.284997	0.155010	0.070358	0.205294

[24]:
```
import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method= 'ward'))
```



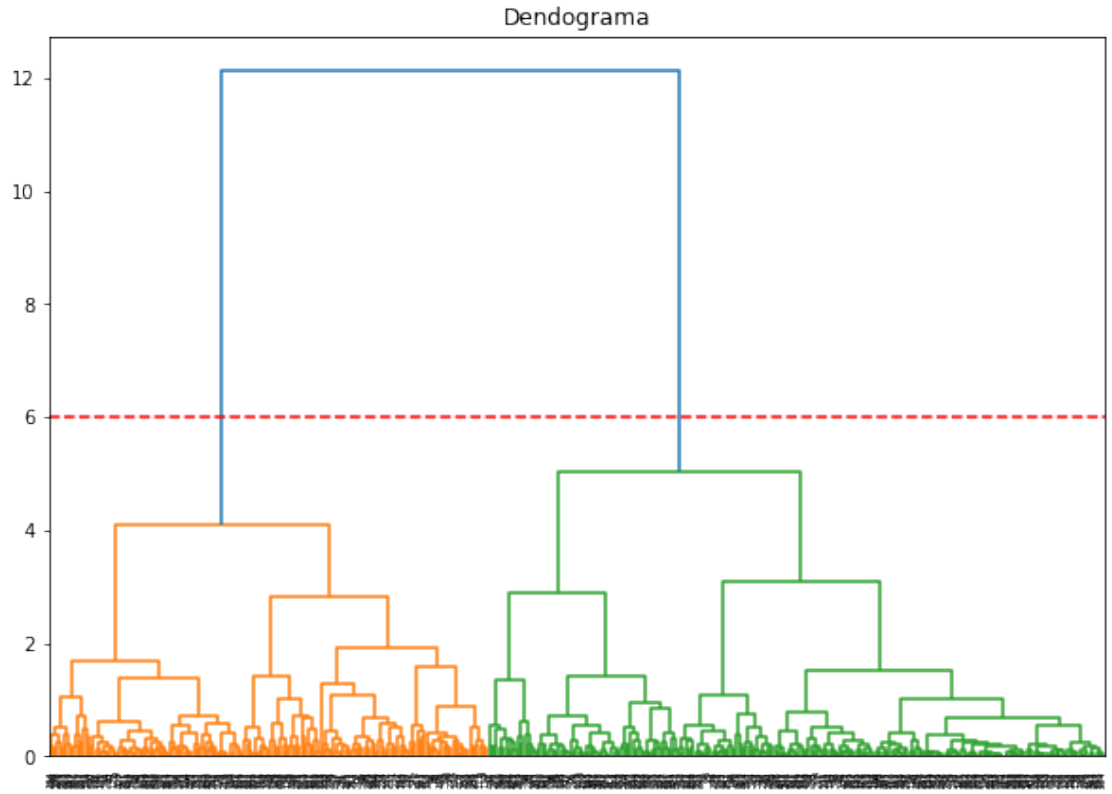
[25]:
```
print(dend['color_list'])
```

```
['C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
```

```
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
```

```
plt.axhline(y=6, color = 'r', linestyle = '--')
```

[29]: <matplotlib.lines.Line2D at 0x735ac5b7e430>



[30]:
```
data
```

[30]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	
3	1	3	13265	1196	4221	6404	507	
4	2	3	22615	5410	7198	3915	1777	
..	…	…	…	…	…	…	…	
435	1	3	29703	12051	16027	13135	182	
436	1	3	39228	1431	764	4510	93	
437	2	3	14531	15488	30243	437	14841	
438	1	3	10290	1981	2232	1038	168	
439	1	3	2787	1698	2510	65	477	

```
2          7844
3          1788
4          5185
..          …
435        2204
436        2346
437        1867
438        2125
439          52

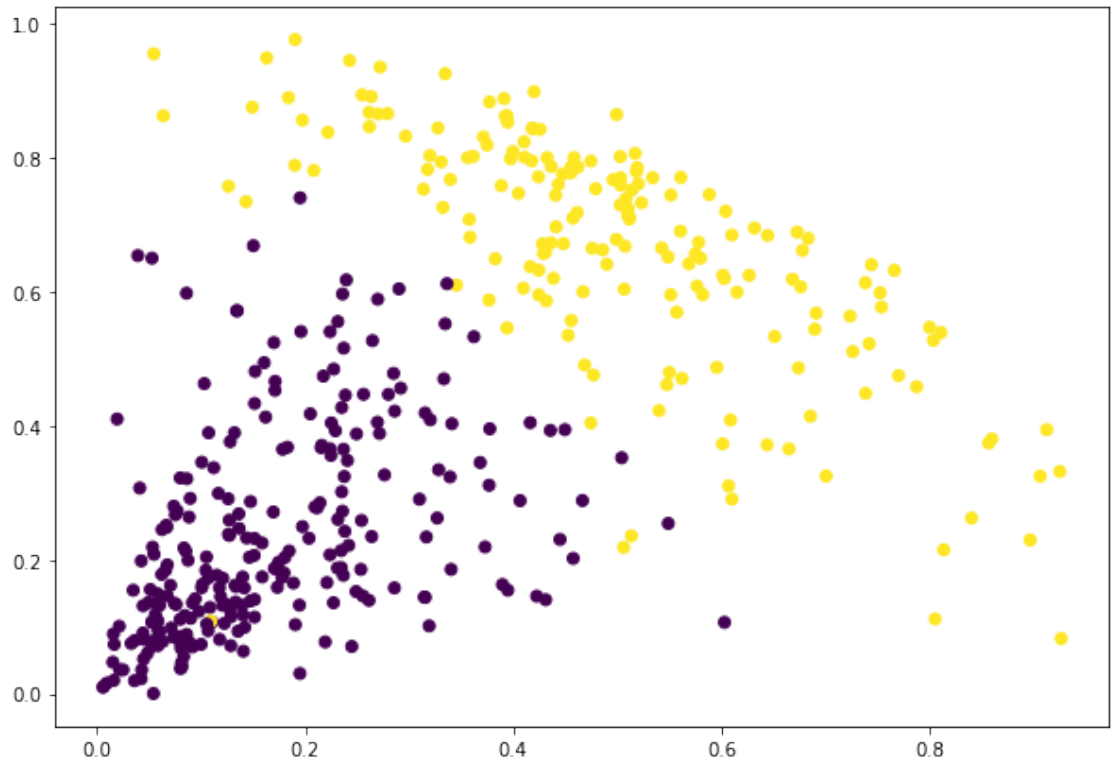
[440 rows x 8 columns]
```

[31]:
```
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters = 2, affinity='euclidean', linkage␣
 ↪= 'ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

[31]:
```
array([1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,
       0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1,
       1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 1, 0,
       0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1,
       0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0,
       0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1,
       0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1,
       0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
       0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 0,
       0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1,
       1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1,
       1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0,
       1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1,
       1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1])
```

[33]:
```
plt.figure(figsize=(10,7))
plt.scatter(data_scaled['Milk'], data_scaled['Grocery'], c = cluster.labels_)
```

[33]:
```
<matplotlib.collections.PathCollection at 0x735ac291ac10>
```



[34]:
```
#SEPARACIÓN DE GRUPOS MEDIANTE PCA
campos = data_scaled.values
print(campos)
```

```
[[1.11821406e-04 1.67732109e-04 7.08332695e-01 … 1.19648904e-02
  1.49505220e-01 7.48085205e-02]
 [1.25321880e-04 1.87982820e-04 4.42198253e-01 … 1.10408576e-01
  2.06342475e-01 1.11285829e-01]
 [1.24839188e-04 1.87258782e-04 3.96551681e-01 … 1.50119124e-01
  2.19467293e-01 4.89619296e-01]
 …
 [5.01633106e-05 7.52449659e-05 3.64461533e-01 … 1.09606834e-02
  3.72236846e-01 4.68274505e-02]
 [9.11309417e-05 2.73392825e-04 9.37737390e-01 … 9.45939175e-02
  1.53099982e-02 1.93653251e-01]
 [2.41225630e-04 7.23676891e-04 6.72295832e-01 … 1.56796660e-02
  1.15064626e-01 1.25437328e-02]]
```

[35]:
```
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

```
[35]: array([[ 3.40370496e-02, -1.80420878e-01],
             [ 3.51057368e-01, -4.66441613e-02],
             [ 3.05375430e-01,  5.55510662e-02],
             [-3.83677396e-01,  1.61789121e-01],
             [-3.18591263e-01, -7.69700156e-02],
             [ 3.73478459e-02, -1.19851588e-01],
             [-1.20176243e-01, -2.43275779e-01],
             [ 2.39772413e-01, -1.04149634e-01],
             [ 1.65698845e-01, -1.87946102e-01],
             [ 5.68883753e-01, -1.09187885e-01],
             [ 5.50490278e-01,  8.66514506e-02],
             [-3.80404330e-01, -1.63301699e-01],
             [-2.16997115e-01, -2.19645753e-01],
             [-4.15793549e-02, -1.73397544e-01],
             [-1.33933983e-01, -2.36377213e-01],
             [-3.35298752e-01, -2.30915370e-01],
             [ 7.26283916e-01, -7.33583822e-02],
             [ 5.14896222e-02,  7.05034152e-03],
             [-1.39371504e-01, -1.51057486e-01],
             [ 1.64608998e-01, -2.12412480e-01],
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             [-3.23995148e-01,  2.39147842e-01],
             [-4.98705749e-01,  4.14682390e-02],
             [ 2.12232953e-01,  5.98882720e-03],
             [-6.30009465e-02, -1.35347961e-01],
             [-1.79665317e-01, -2.61548271e-01],
             [-4.14823639e-01,  3.42279073e-02],
             [-4.79249014e-01, -2.07953429e-01],
             [ 6.73129427e-01, -4.79248386e-02],
             [-5.70961913e-01, -1.91841970e-01],
             [-1.67202588e-01, -2.10793120e-01],
             [ 2.95611870e-01,  2.02389171e-01],
             [-5.17167853e-01, -2.21203305e-01],
             [-4.24196938e-01, -5.27010145e-02],
             [ 3.54771268e-01, -3.65412400e-02],
             [ 7.27927338e-01, -5.33108689e-02],
             [-4.52523392e-01, -1.65167176e-01],
             [ 1.29597534e-01, -1.53518969e-01],
             [ 6.18731741e-01, -9.98567672e-02],
             [-6.19228600e-01, -4.13006443e-02],
             [-4.13793874e-01,  1.28551576e-01],
             [-3.43063809e-01, -1.56418846e-01],
             [ 3.07519786e-01, -2.28921480e-01],
             [ 7.53223768e-01, -8.39338939e-02],
             [ 2.24695460e-01, -2.01776979e-01],
             [ 6.22063312e-01, -1.57586831e-02],
             [ 6.79846268e-01, -4.73480600e-02],
```

```
[ 3.27795558e-01, -9.55443701e-02],
[ 1.44410163e-01, -2.16552831e-01],
[ 6.74077008e-01, -6.84096223e-02],
[-3.15602398e-01,  2.61471315e-01],
[ 4.43127098e-01, -1.51451615e-01],
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[-5.70637048e-01, -1.32305756e-01],
[ 1.61897456e-01, -1.02412575e-01],
[ 6.80069130e-01, -1.66615640e-02],
[ 5.03231325e-01, -1.48638052e-01],
[-5.01834682e-01, -1.78653901e-01],
[ 3.05604240e-01, -1.91464732e-01],
[ 1.00890901e-01, -2.67697600e-01],
[ 3.99561074e-01, -1.65831759e-01],
[ 1.40236863e-01,  8.28181774e-02],
[ 4.08870795e-01, -7.09228640e-02],
[-9.03889130e-02,  2.64962008e-01],
[ 7.69990379e-01, -1.09298565e-01],
[ 7.35890990e-01, -1.16366144e-01],
[-4.81786081e-02, -2.21148988e-01],
[ 3.25118853e-01,  4.99184692e-01],
[-3.82260364e-01, -1.29126685e-01],
[-4.34980462e-01,  3.00208993e-01],
[ 7.45489538e-02, -4.11025454e-02],
[ 4.17519668e-02,  6.42719663e-01],
[-2.62294212e-01,  8.45335699e-02],
[ 2.26029453e-01, -1.18599620e-01],
[-6.11228251e-01, -4.80484709e-04],
[ 4.29197018e-01,  6.15604204e-01],
[ 4.81706738e-01, -1.79409872e-01],
[-4.75041077e-01, -6.15378670e-02],
[ 4.76281160e-01,  4.24517554e-02],
[-2.98067142e-01, -2.11883345e-01],
[ 7.67546282e-01, -7.70688157e-02],
[ 2.30031687e-01, -1.54241186e-01],
[-5.50753086e-01, -1.17615669e-01],
[-1.94789047e-01, -1.90991741e-01],
[ 6.64313844e-01, -1.34824110e-01],
[ 4.54220988e-01, -2.80065613e-02],
[-4.38157051e-01, -7.06239740e-02],
[-2.33191454e-01,  5.23510639e-01],
[-4.38507986e-01, -1.00163161e-01],
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[-3.84725272e-01,  3.24407654e-01],
[ 6.09426658e-01, -5.47808736e-02],
[-1.66045468e-01,  8.67796686e-01],
```

```
[ 5.02165348e-01, -1.12823857e-01],
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[ 7.55122640e-01, -1.08955508e-01],
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[ 1.06014532e-01,  4.87077633e-01],
[-4.01264649e-01,  2.60610397e-01],
[ 2.18121540e-01, -8.19292232e-02],
[ 7.16631603e-01, -2.25380234e-02],
[ 5.29503567e-01,  6.70270334e-02],
[-5.07251706e-01,  5.60710953e-02],
[-4.01188893e-01, -1.64911406e-01],
[-4.47945893e-01, -1.86856753e-01],
[ 6.92420411e-01, -1.12234792e-01],
[ 4.20871074e-01, -8.95874663e-02],
[ 6.35552091e-01, -2.48684459e-02],
[ 7.04382875e-01, -4.14433062e-02],
[-4.87851159e-01, -3.87639346e-02],
[ 3.22221124e-01, -1.79709017e-01],
[-4.73638478e-01,  1.90964168e-01],
[-4.89434312e-01, -1.29715397e-02],
[-5.45415376e-01, -8.96347007e-02],
[-4.40321505e-01, -3.02049954e-02],
[-3.59736185e-01, -1.20462857e-01],
[ 4.68667080e-03, -5.37226553e-02],
[-5.24106450e-01,  1.92511056e-02],
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[-4.87069263e-01, -1.06634112e-01],
[-1.03450303e-01,  2.35158316e-01],
[-6.14165133e-01, -5.35381996e-02],
[ 1.29981880e-01, -6.24571595e-02],
[-5.67154960e-01, -9.65112949e-02],
[-5.62544329e-01, -2.43749455e-02],
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[-1.69405331e-01, -2.07508734e-01],
[ 6.34852210e-01,  7.58407518e-02],
[-5.99064164e-01, -1.72225497e-01],
[-3.59331379e-01,  4.57567083e-01],
[-4.18374199e-01, -5.44280241e-02],
[-4.63713323e-01, -1.99604919e-01],
[-5.05456383e-01, -1.81808278e-01],
[-3.52285781e-01, -1.41640494e-01],
[-2.99690179e-01, -9.28211461e-02],
[ 3.63475981e-01,  2.16338091e-01],
[ 5.19735352e-01, -3.69874861e-02],
[-2.60365473e-01, -1.97639995e-01],
[ 5.97389341e-02, -6.20719662e-02],
[-2.70392295e-01, -1.30816527e-01],
```

```
[-3.38769193e-01, -1.73375038e-01],
[-4.25273239e-01, -1.57577342e-01],
[-4.34514025e-01,  4.15564262e-01],
[-4.82110726e-01, -2.06866073e-01],
[ 2.96333426e-01, -2.43962756e-01],
[-3.63983846e-01, -1.22613473e-01],
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[ 3.94004028e-01,  9.95566822e-02],
[-5.06978600e-01, -9.04014501e-02],
[ 2.86859551e-01,  8.79092496e-02],
[-4.68027517e-01, -1.31938521e-01],
[ 7.12664171e-01, -1.12878705e-01],
[ 5.84135780e-01, -1.48282797e-01],
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[ 5.86657814e-01, -6.50949747e-02],
[ 7.57866799e-01, -9.00865812e-02],
[ 6.49420759e-01, -8.90239974e-02],
[-5.90165441e-01, -1.08852374e-01],
[-4.30908934e-01, -8.27549896e-02],
[ 6.44506232e-01, -8.10786344e-02],
[ 4.01301513e-01,  1.04069927e-02],
[ 2.67136265e-01, -7.41762676e-02],
[ 4.53894565e-01, -8.54217066e-02],
[ 2.92116091e-01, -1.14922431e-01],
[-5.00801129e-01, -2.83747021e-03],
[-3.31542329e-01,  1.22155150e-01],
[ 7.59204668e-01, -3.36936733e-02],
[ 7.25036896e-01,  2.35040300e-02],
[ 6.88544230e-01, -5.11999893e-02],
[ 7.45287159e-01, -8.62137068e-02],
[ 5.56620944e-01,  3.12251305e-01],
[ 6.54122938e-01, -1.07705750e-01],
[-4.85880585e-01, -8.07793100e-02],
[-1.83084867e-01, -6.67240374e-02],
[-1.14072184e-01, -1.08654684e-01],
[ 1.42664711e-01,  4.75047756e-01],
[-7.02023479e-03, -1.95836482e-01],
[-4.20557019e-01, -8.00556892e-02],
[ 7.34681336e-01, -6.37764400e-02],
[ 5.57753564e-02,  3.86989624e-01],
[ 6.51175951e-01, -1.03707929e-01],
[-2.11688710e-01,  6.72622227e-02],
[-2.45868911e-01,  5.08336716e-01],
[ 5.31899526e-01,  2.93475419e-01],
```

```
[ 3.49267208e-01, -7.08877753e-02],
[ 7.18728186e-01, -2.26513597e-02],
[-4.03482871e-01,  1.70807142e-01],
[-5.65892127e-01, -1.65077097e-01],
[-2.72364466e-01, -1.56853473e-01],
[ 7.15311782e-01, -8.35088481e-02],
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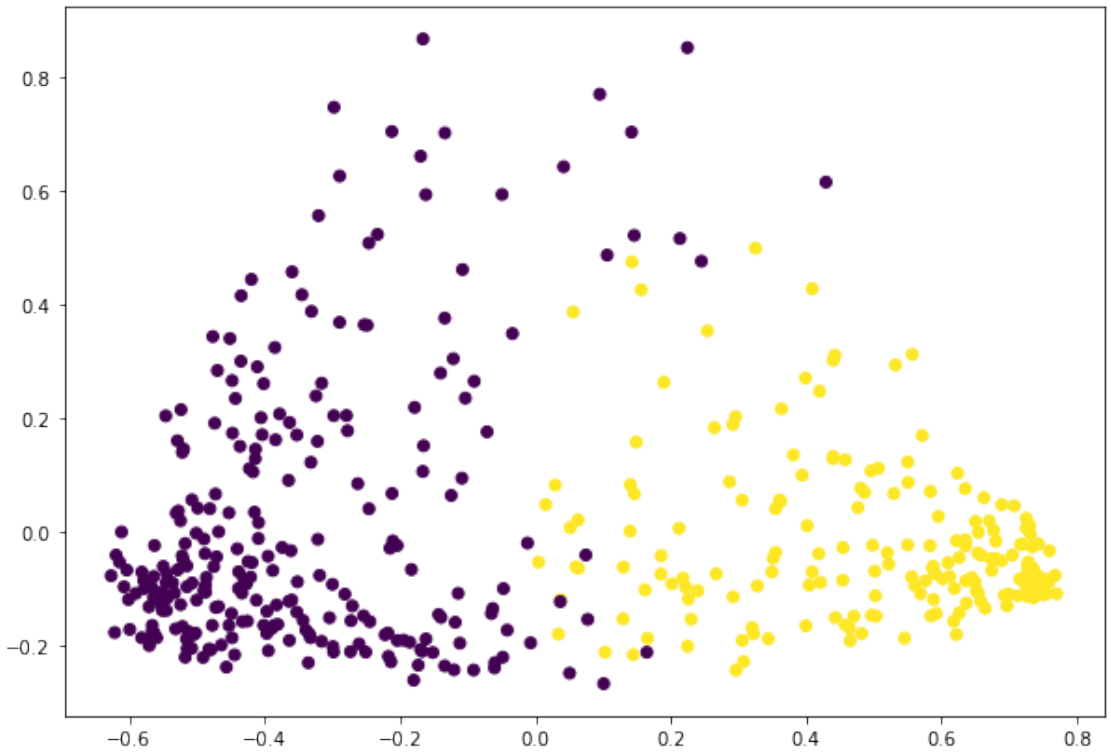
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[36]:
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plt.figure(figsize = (10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

[36]:
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[37]:
```
dataframe = pd.DataFrame(grupos, columns= ['grupo'])
dataframe
```

[37]:

	grupo
0	1
1	1
2	1
3	0
4	0
..	…
435	0
436	0
437	1
438	0
439	1

[440 rows x 1 columns]

[41]:
```
dataframe2 = pd.concat([data, dataframe], axis= 1, join = 'inner')
dataframe2
```

[41]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	\
0	2	3	12669	9656	7561	214	2674	
1	2	3	7057	9810	9568	1762	3293	
2	2	3	6353	8808	7684	2405	3516	
3	1	3	13265	1196	4221	6404	507	
4	2	3	22615	5410	7198	3915	1777	
5	2	3	9413	8259	5126	666	1795	
6	2	3	12126	3199	6975	480	3140	
7	2	3	7579	4956	9426	1669	3321	
8	1	3	5963	3648	6192	425	1716	
9	2	3	6006	11093	18881	1159	7425	
10	2	3	3366	5403	12974	4400	5977	
11	2	3	13146	1124	4523	1420	549	
12	2	3	31714	12319	11757	287	3881	
13	2	3	21217	6208	14982	3095	6707	
14	2	3	24653	9465	12091	294	5058	
15	1	3	10253	1114	3821	397	964	
16	2	3	1020	8816	12121	134	4508	
17	1	3	5876	6157	2933	839	370	
18	2	3	18601	6327	10099	2205	2767	
19	1	3	7780	2495	9464	669	2518	
20	2	3	17546	4519	4602	1066	2259	
21	1	3	5567	871	2010	3383	375	
22	1	3	31276	1917	4469	9408	2381	
23	2	3	26373	36423	22019	5154	4337	
24	2	3	22647	9776	13792	2915	4482	

25	2	3	16165	4230	7595	201	4003
26	1	3	9898	961	2861	3151	242
27	1	3	14276	803	3045	485	100
28	2	3	4113	20484	25957	1158	8604
29	1	3	43088	2100	2609	1200	1107
30	1	3	18815	3610	11107	1148	2134
31	1	3	2612	4339	3133	2088	820
32	1	3	21632	1318	2886	266	918
33	1	3	29729	4786	7326	6130	361
34	1	3	1502	1979	2262	425	483
35	2	3	688	5491	11091	833	4239
36	1	3	29955	4362	5428	1729	862
37	2	3	15168	10556	12477	1920	6506
38	2	3	4591	15729	16709	33	6956
39	1	3	56159	555	902	10002	212
40	1	3	24025	4332	4757	9510	1145
41	1	3	19176	3065	5956	2033	2575
42	2	3	10850	7555	14961	188	6899
43	2	3	630	11095	23998	787	9529
44	2	3	9670	7027	10471	541	4618
45	2	3	5181	22044	21531	1740	7353
46	2	3	3103	14069	21955	1668	6792
47	2	3	44466	54259	55571	7782	24171
48	2	3	11519	6152	10868	584	5121
49	2	3	4967	21412	28921	1798	13583
50	1	3	6269	1095	1980	3860	609
51	1	3	3347	4051	6996	239	1538
52	2	3	40721	3916	5876	532	2587
53	2	3	491	10473	11532	744	5611
54	1	3	27329	1449	1947	2436	204
55	1	3	5264	3683	5005	1057	2024
56	2	3	4098	29892	26866	2616	17740
57	2	3	5417	9933	10487	38	7572
58	1	3	13779	1970	1648	596	227
59	1	3	6137	5360	8040	129	3084
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61	2	3	35942	38369	59598	3254	26701
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63	2	3	9396	11601	15775	2896	7677
64	1	3	4760	1227	3250	3724	1247
65	2	3	85	20959	45828	36	24231
66	1	3	9	1534	7417	175	3468
67	2	3	19913	6759	13462	1256	5141
68	1	3	2446	7260	3993	5870	788
69	1	3	8352	2820	1293	779	656
70	1	3	16705	2037	3202	10643	116
71	1	3	18291	1266	21042	5373	4173

72	1	3	4420	5139	2661	8872	1321
73	2	3	19899	5332	8713	8132	764
74	2	3	8190	6343	9794	1285	1901
75	1	3	20398	1137	3	4407	3
76	1	3	717	3587	6532	7530	529
77	2	3	12205	12697	28540	869	12034
78	1	3	10766	1175	2067	2096	301
79	1	3	1640	3259	3655	868	1202
80	1	3	7005	829	3009	430	610
81	2	3	219	9540	14403	283	7818
82	2	3	10362	9232	11009	737	3537
83	1	3	20874	1563	1783	2320	550
84	2	3	11867	3327	4814	1178	3837
85	2	3	16117	46197	92780	1026	40827
86	2	3	22925	73498	32114	987	20070
87	1	3	43265	5025	8117	6312	1579
88	1	3	7864	542	4042	9735	165
89	1	3	24904	3836	5330	3443	454
90	1	3	11405	596	1638	3347	69
91	1	3	12754	2762	2530	8693	627
92	2	3	9198	27472	32034	3232	18906
93	1	3	11314	3090	2062	35009	71
94	2	3	5626	12220	11323	206	5038
95	1	3	3	2920	6252	440	223
96	2	3	23	2616	8118	145	3874
97	1	3	403	254	610	774	54
98	1	3	503	112	778	895	56
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138	3271	0
139	929	1
140	2616	0
141	1450	0
142	3	0
143	318	0
144	201	0
145	4430	1
146	520	0

147	526	0
148	434	0
149	1440	0
150	1067	0
151	1774	1
152	184	0
153	1627	1
154	8	0
155	2153	1
156	3182	1
157	418	0
158	1682	1
159	303	1
160	2157	1
161	2233	0
162	610	0
163	446	1
164	238	1
165	2379	1
166	3637	1
167	693	1
168	429	0
169	750	0
170	323	1
171	6250	1
172	707	1
173	716	1
174	1442	1
175	1697	1
176	230	0
177	2631	0
178	395	0
179	2165	1
180	2794	0
181	8550	0
182	156	1
183	47943	1
184	11	1
185	247	0
186	727	0
187	3	1
188	404	1
189	1856	1
190	64	0
191	84	0
192	409	0
193	666	1

194	1142	0
195	1755	0
196	2876	0
197	1468	1
198	697	0
199	347	0
200	731	1
201	1681	1
202	6854	0
203	18	1
204	1328	1
205	710	1
206	285	0
207	120	1
208	806	1
209	797	1
210	2100	0
211	2870	1
212	1775	0
213	1215	1
214	791	1
215	2388	1
216	674	1
217	1158	0
218	6372	1
219	130	0
220	749	0
221	239	1
222	375	0
223	1360	0
224	659	0
225	786	0
226	1553	0
227	689	0
228	203	1
229	980	0
230	2137	0
231	490	1
232	834	0
233	7	1
234	2563	0
235	295	1
236	225	0
237	1215	0
238	216	0
239	2253	0
240	2564	0

241	1047	0
242	578	0
243	2398	0
244	1970	1
245	2784	1
246	610	0
247	659	0
248	572	0
249	291	0
250	710	0
251	5121	1
252	1693	0
253	1391	0
254	3265	1
255	615	0
256	373	0
257	987	0
258	3321	0
259	818	0
260	548	0
261	287	0
262	655	0
263	411	1
264	1265	1
265	3636	1
266	2563	1
267	3628	0
268	698	1
269	204	0
270	56	0
271	1550	1
272	1040	1
273	1824	0
274	1153	1
275	379	1
276	2503	0
277	139	0
278	1409	0
279	1721	0
280	1104	1
281	2079	0
282	1404	0
283	1384	0
284	2406	0
285	18	0
286	128	0
287	1027	0

288	258	0
289	22	0
290	1522	1
291	686	0
292	1060	1
293	741	1
294	1854	0
295	254	0
296	898	0
297	531	0
298	1037	1
299	259	1
300	2005	0
301	172	1
302	555	1
303	59	1
304	2410	1
305	211	1
306	1543	1
307	925	0
308	656	0
309	806	1
310	1117	0
311	117	0
312	142	1
313	297	0
314	1233	0
315	3508	1
316	1059	0
317	1637	1
318	51	0
319	1625	1
320	834	1
321	1113	0
322	229	0
323	573	0
324	1092	0
325	5609	0
326	834	0
327	522	1
328	1534	0
329	739	0
330	1043	0
331	1102	1
332	2602	0
333	1215	1
334	3486	0

335	2139	0
336	778	0
337	868	0
338	550	0
339	1942	0
340	1371	1
341	2158	1
342	1328	1
343	37	1
344	379	0
345	303	1
346	1115	1
347	1022	0
348	665	1
349	445	1
350	995	0
351	3137	1
352	195	1
353	1111	1
354	2341	0
355	127	1
356	548	0
357	110	1
358	4100	1
359	776	1
360	503	0
361	405	0
362	712	1
363	314	0
364	468	0
365	3105	1
366	447	0
367	342	0
368	558	0
369	296	0
370	2235	0
371	790	0
372	4829	1
373	3113	0
374	686	0
375	70	0
376	1426	1
377	1242	0
378	1114	1
379	179	1
380	270	0
381	532	0

382	2893	0
383	361	0
384	5120	1
385	1068	0
386	967	1
387	961	0
388	406	0
389	684	0
390	1000	0
391	1827	1
392	654	1
393	819	0
394	452	0
395	290	0
396	2213	1
397	743	0
398	247	0
399	375	0
400	1014	0
401	1902	0
402	340	0
403	288	0
404	715	0
405	378	0
406	960	0
407	553	1
408	344	0
409	5137	0
410	1892	0
411	4365	1
412	62	1
413	2435	0
414	290	0
415	1874	1
416	993	1
417	1063	1
418	776	1
419	1521	0
420	1393	1
421	1784	0
422	1218	0
423	668	0
424	249	0
425	1886	0
426	1894	1
427	1163	0
428	317	1

```
429        2501      0
430        2080      1
431        1498      0
432         395      0
433        1449      1
434         838      0
435        2204      0
436        2346      0
437        1867      1
438        2125      0
439          52      1
```

[40]:
```
pd.options.display.max_rows = None
```

[42]:
```
#Análisis del caso de estudiantes
data = pd.DataFrame(np.array([[7,6.5,9.2,8.6,8],
                              [7.5,9.4,7.3,7,7],
                              [7.6,9.2,8,8,7.5],
                              [5,6.5,6.5,7,9],
                              [6,6,7.8,8.9,7.3],
                              [7.8,9.6,7.7,8,6.5],
                              [6.3,6.4,8.2,9,7.2],
                              [7.9,9.7,7.5,8,6],
                              [6,6,6.5,5.5,8.7],
                              [6.8,7.2,8.7,9,7]]),
                 columns = ['Matematicas', 'Ciencias', 'Español',␣
 ↪'Historia','EdFisica'])

data
```

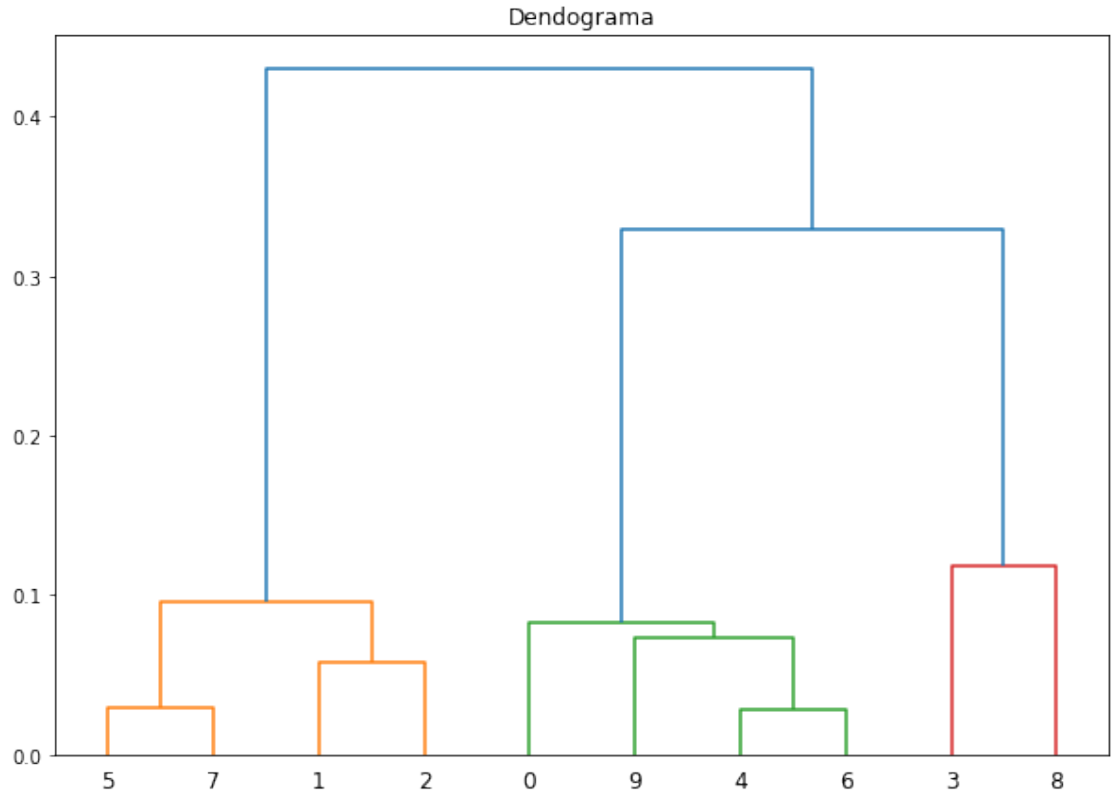
[42]:

	Matematicas	Ciencias	Español	Historia	EdFisica
0	7.0	6.5	9.2	8.6	8.0
1	7.5	9.4	7.3	7.0	7.0
2	7.6	9.2	8.0	8.0	7.5
3	5.0	6.5	6.5	7.0	9.0
4	6.0	6.0	7.8	8.9	7.3
5	7.8	9.6	7.7	8.0	6.5
6	6.3	6.4	8.2	9.0	7.2
7	7.9	9.7	7.5	8.0	6.0
8	6.0	6.0	6.5	5.5	8.7
9	6.8	7.2	8.7	9.0	7.0

[43]:
```
from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

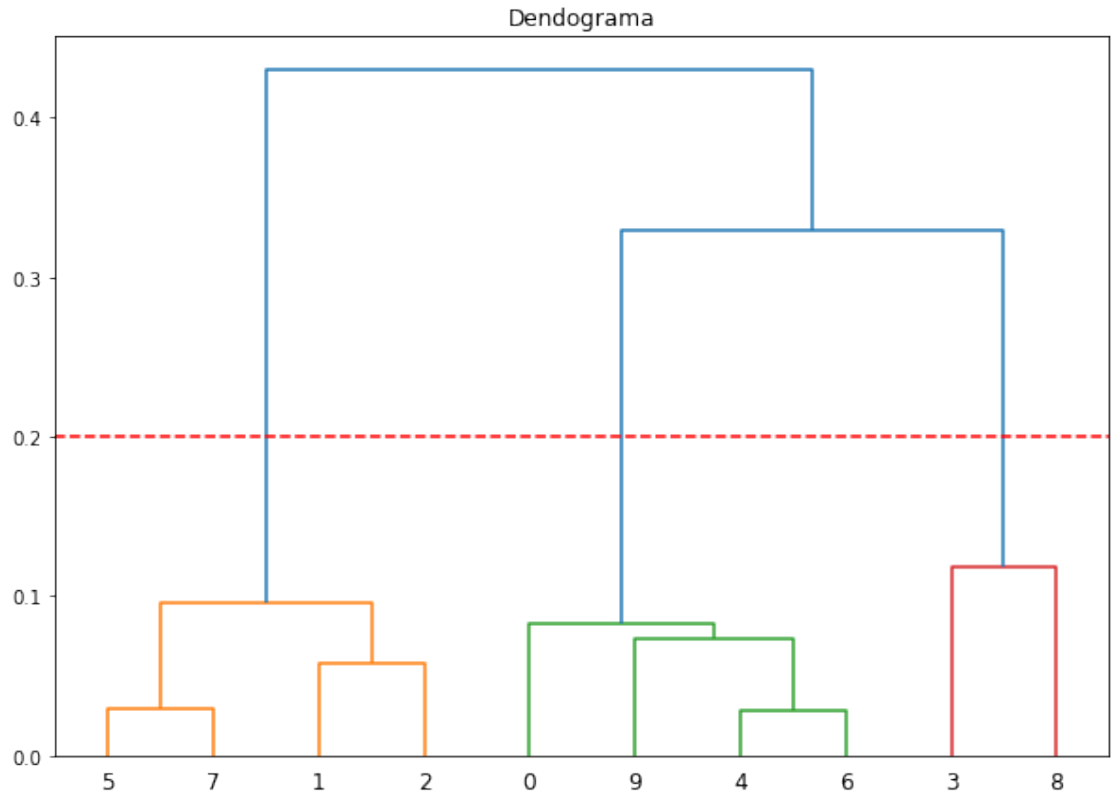
[43]:
```
array([[0.39512724, 0.36690387, 0.51931009, 0.48544204, 0.45157399],
       [0.4360023 , 0.54645622, 0.42437557, 0.40693548, 0.40693548],
       [0.42050606, 0.50903365, 0.44263796, 0.44263796, 0.41497309],
       [0.32308533, 0.42001094, 0.42001094, 0.45231947, 0.5815536 ],
       [0.36834085, 0.36834085, 0.47884311, 0.54637226, 0.44814804],
       [0.43703137, 0.53788477, 0.43142841, 0.44823731, 0.36419281],
       [0.37600718, 0.38197555, 0.48940617, 0.53715311, 0.42972249],
       [0.44671284, 0.54849551, 0.42409446, 0.45236743, 0.33927557],
       [0.40434535, 0.40434535, 0.4380408 , 0.37064991, 0.58630076],
       [0.39015437, 0.41310462, 0.49916809, 0.51638078, 0.4016295 ]])
```

[44]:
```
import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method= 'ward'))
```



[45]:
```
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method = 'ward'))
plt.axhline(y=0.2, color = 'r', linestyle = '--')
```

[45]: <matplotlib.lines.Line2D at 0x735ac07e0160>

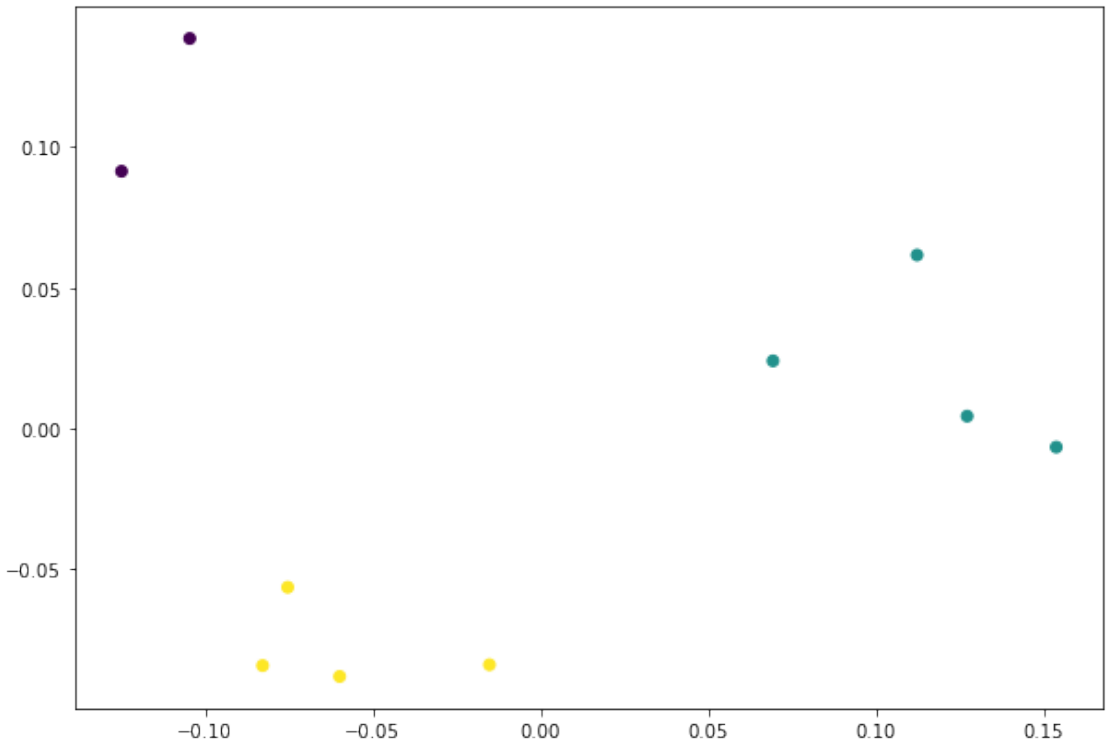


[46]:
```
#Analisis grafico con PCA
campos = data_scaled
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

[46]:
```
array([[-0.07537384, -0.05649076],
       [ 0.11233306,  0.06156361],
       [ 0.06937082,  0.02395785],
       [-0.12493186,  0.09133353],
       [-0.08285196, -0.08431667],
       [ 0.12725973,  0.00431553],
       [-0.05987387, -0.08818057],
       [ 0.15385473, -0.00668172],
       [-0.1045932 ,  0.13852098],
       [-0.01519363, -0.08402179]])
```

[48]:
```
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters = 3, affinity='euclidean', linkage␣
 ↪= 'ward')
grupos = cluster.fit_predict(data_scaled)
grupos
plt.figure(figsize = (10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

[48]:
```
<matplotlib.collections.PathCollection at 0x735ac4b017f0>
```



[49]:
```
dataframe = pd.DataFrame(grupos, columns= ['grupo'])
dataframe
dataframe2 = pd.concat([data, dataframe], axis= 1, join = 'inner')
dataframe2
```

[49]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
0	7.0	6.5	9.2	8.6	8.0	2
1	7.5	9.4	7.3	7.0	7.0	1
2	7.6	9.2	8.0	8.0	7.5	1
3	5.0	6.5	6.5	7.0	9.0	0
4	6.0	6.0	7.8	8.9	7.3	2
5	7.8	9.6	7.7	8.0	6.5	1
6	6.3	6.4	8.2	9.0	7.2	2

7	7.9	9.7	7.5	8.0	6.0	1
8	6.0	6.0	6.5	5.5	8.7	0
9	6.8	7.2	8.7	9.0	7.0	2

[50]:
```
dataframe2.index = ['Lucia', 'Pedro', 'Ines','Luis','Andres', 'Ana', 'Carlos',␣
↪'Jose', 'Sonia', 'Maria']
dataframe2
```

[50]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
Lucia	7.0	6.5	9.2	8.6	8.0	2
Pedro	7.5	9.4	7.3	7.0	7.0	1
Ines	7.6	9.2	8.0	8.0	7.5	1
Luis	5.0	6.5	6.5	7.0	9.0	0
Andres	6.0	6.0	7.8	8.9	7.3	2
Ana	7.8	9.6	7.7	8.0	6.5	1
Carlos	6.3	6.4	8.2	9.0	7.2	2
Jose	7.9	9.7	7.5	8.0	6.0	1
Sonia	6.0	6.0	6.5	5.5	8.7	0
Maria	6.8	7.2	8.7	9.0	7.0	2

[ ]:
```

```